

ASSIGNMENT #06

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, April 17th TA's Mailbox

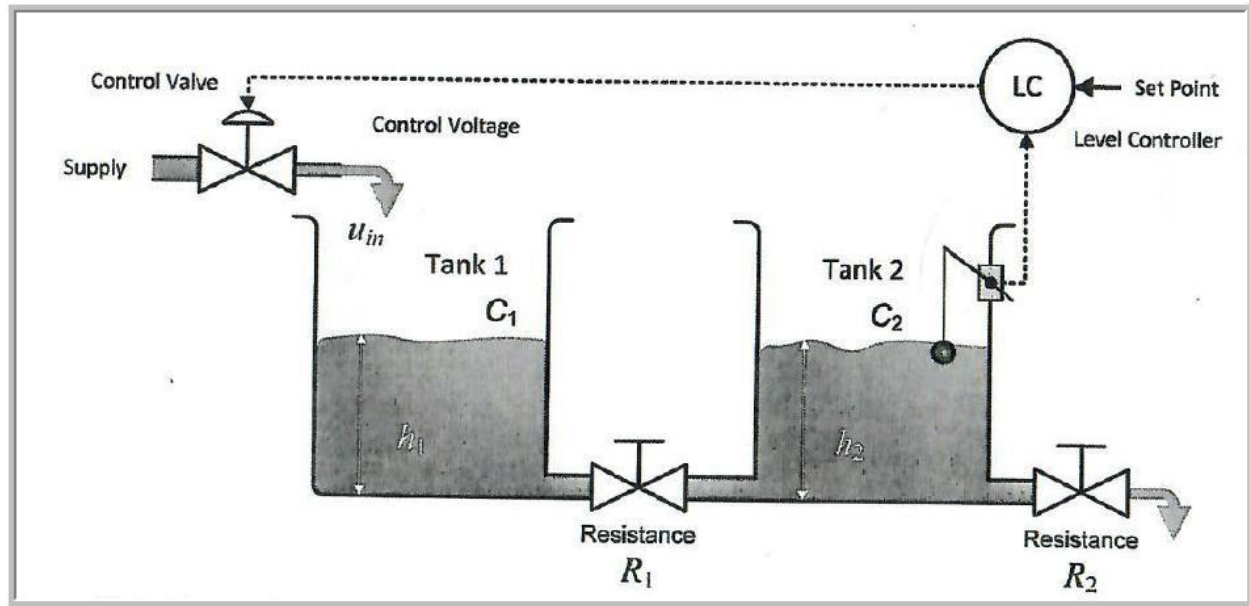
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name in the report.
 - Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
 - Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
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Problem: Two-Tank Level Control System

In many chemical process applications, cascaded tank systems are used to attenuate disturbances and improve level regulation. Consider the two-tank system shown below, where liquid flows from Tank 1 into Tank 2.



A level transmitter is installed in Tank 2 and provides feedback to a PID controller, which manipulates the inlet flow rate to regulate the level in Tank 2.

The capacitance of the first and second tank is $C_1=C_2=1.0 \text{ m}^3/\text{m}$ and the outlet resistances of the first and second tank $R_1=0.5 \text{ m}/(\text{m}^3/\text{min})$ and $R_2=0.333 \text{ m}/(\text{m}^3/\text{min})$. The volumetric flow rate through the gate valves varies linearly with the pressure drop. Assume that the time delays in the control loop including the transportation lag between the first and second tank amount to around 35% of the dominant time constant of the system.

1. Derive the process and disturbance transfer functions assuming that the inlet flow acts as the manipulated variable (MV) and also as the disturbance variable (DV).
2. Design a PID controller with $\text{GM}=3$ and $\text{PM}=60$ degrees and perform a simulation for set point and disturbance change of unit size.
3. Compare your controller with:
 - a. Ziegler-Nichols Open Loop
 - b. Cohen-Coon
 - c. Ziegler-Nichols Closed Loop
 - d. Luyben-Tyreus
 - e. IMC-based tuning
 - f. Your own design

Chemical Process Control

Present your results in a tabular form and compute the following metrics for both set point tracking and disturbance rejection of unit size.

1. Overshoot (if present)
2. Settling time
3. Gain Margin
4. Phase Margin
5. Maximum amount of time delay allowed before the onset of instability
6. Peak value M_s of the sensitivity function
7. Peak value M_t of the complementary sensitivity function

Include all the simulation graphs in your final report along with the Excel table shown below

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyreus	IMC	My Personal Controller	Selected Controller
Overshoot (if present)								
Settling time								
Gain Margin								
Phase Margin								
Maximum amount of time delay allowed before instability								
Peak Value of the sensitivity function								
Peak Value of the complimentary sensitivity function								
Controller Gain, K_c								
Controller Reset Time, τ_i								
Controller Derivative Time, τ_D								

Discuss the trade-offs observed between performance and robustness for the different controllers. Comment on which controller provides the best overall performance and justify your answer.

Chemical Process Control

Tank mass balances are:

$$C_1 = \frac{dh_1}{dt} = q_{in} - \frac{h_1}{R_1}$$

$$C_2 = \frac{dh_2}{dt} = \frac{h_1}{R_1} - \frac{h_2}{R_2}$$

Laplace transforms of the mass balances are:

$$H_1(s) = \frac{R_1}{R_1 C_1 s + 1} Q_{in}(s)$$

$$H_2(s) = \frac{1/R_1}{C_2 s + 1/R_2} H_1(s)$$

Substituting $H_1(s)$ into $H_2(s)$:

$$H_2(s) = \frac{1/R_1}{C_2 s + 1/R_2} * \frac{R_1}{R_1 C_1 s + 1} Q_{in}(s)$$

Where the process transfer function becomes:

$$G_P(s) = \frac{H_2(s)}{Q_{in}(s)} = \frac{R_2}{(R_1 C_1 s + 1)(R_2 C_2 s + 1)}$$

Or:

$$G_P(s) = \frac{0.333}{(0.5s + 1)(0.333s + 1)}$$

Where the dominant time constant becomes $0.35 * 0.3 = 0.175$.

The process transfer function, including the time delay, is then:

$$G_P(s) = \frac{0.333e^{-0.175s}}{(0.5s + 1)(0.333s + 1)}$$

Since the inlet flow is the disturbance being evaluated, and the disturbance is the same as the dynamic path from the inlet flow to the level in tank two, the disturbance transfer function looks like:

$$G_d(s) = \frac{0.333}{(0.5s + 1)(0.333s + 1)}$$

Then, the following was done in excel (where tau is equal to the square root of tau one multiplied by tau two, and the chi value is tau one plus tau two over the square rooted value previously mentioned).

Chemical Process Control

	A	B	C
1	BODE Stability Criteria Ultimate Gain Analysis	K_p	0.3330
2		τ_p	0.4080
3		θ_p	0.1750
4		ζ	2.0410
5		τ_3	
6		K_c	5.0000
7	OFF Reset All Data ON	τ_I	0.7500
8		τ_D	0.5000
9			
10	AMPLITUDE RATIO AT CRITICAL FREQUENCY	AR_c	0.3251
11			
12	Critical Frequency ($PA=-180^\circ$)	ω_c	12.0905
13	Gain Crossover Frequency ($AR=1$)	ω_g	1.0994
14	Gain Margin	GM	3.0756
15	Phase Margin (deg)	PM	68.9998
16			
17	Ultimate Gain	K_u	15.3782
18	Ultimate Period	P_u	0.5197
19	Maximum Additional Delay	$\Delta\theta_{max}$	1.0954
20		Peak Sensitivity	1.4965
21		Critical Frequency Equation:	0.0000
22		Gain Crossover Frequency Equation:	0.0000
23		Peak Complementary Sensitivity	1.1119

Controller gain and tau I were selected based on trying to get the PM as close to 60 while holding the GM close to 3.

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyreus	IMC	My Personal Controller	Selected Controller
Overshoot (if present)	4%	38%	52%	55%	0%	0.67%	5%	0.67%
Settling time	6	2	3	15	8	5	7	5
Gain Margin	3.0756	1.9716	1.5642	2.3831	4.369	4.0203	3.7443	4.0203
Phase Margin	68.9998	24.9041	13.3927	28.495	64.1909	47.9182	51.5405	47.9182
Maximum amount of time delay allowed before instability	1.0954	0.1468	0.0723	0.1949	0.725	0.487	0.4977	0.487
Peak Value of the sensitivity function	1.4965	2.8686	4.902	2.4349	1.4417	1.5851	1.5854	1.5851
Peak Value of the complimentary sensitivity function	1.1119	2.3914	4.4897	2.0433	1.002	1.2558	1.1693	1.2558
Controller Gain, K_c	5	12.87	13.14	10.48	7.28	7.15	8	7.15
Controller Reset Time, τ_I	0.75	0.583	0.42	0.564	1.489	0.833	1	0.833
Controller Derivative Time, τ_D	0.5	0	0	0	0	0	0	0

The IMC has the best performance with essentially no overshoot and a quick settling time. Luyben-Tyreus also is a strong candidate for a good controller, as it has no overshoot but a slightly slower settling time.

Chemical Process Control
GM=3, PM=60 deg



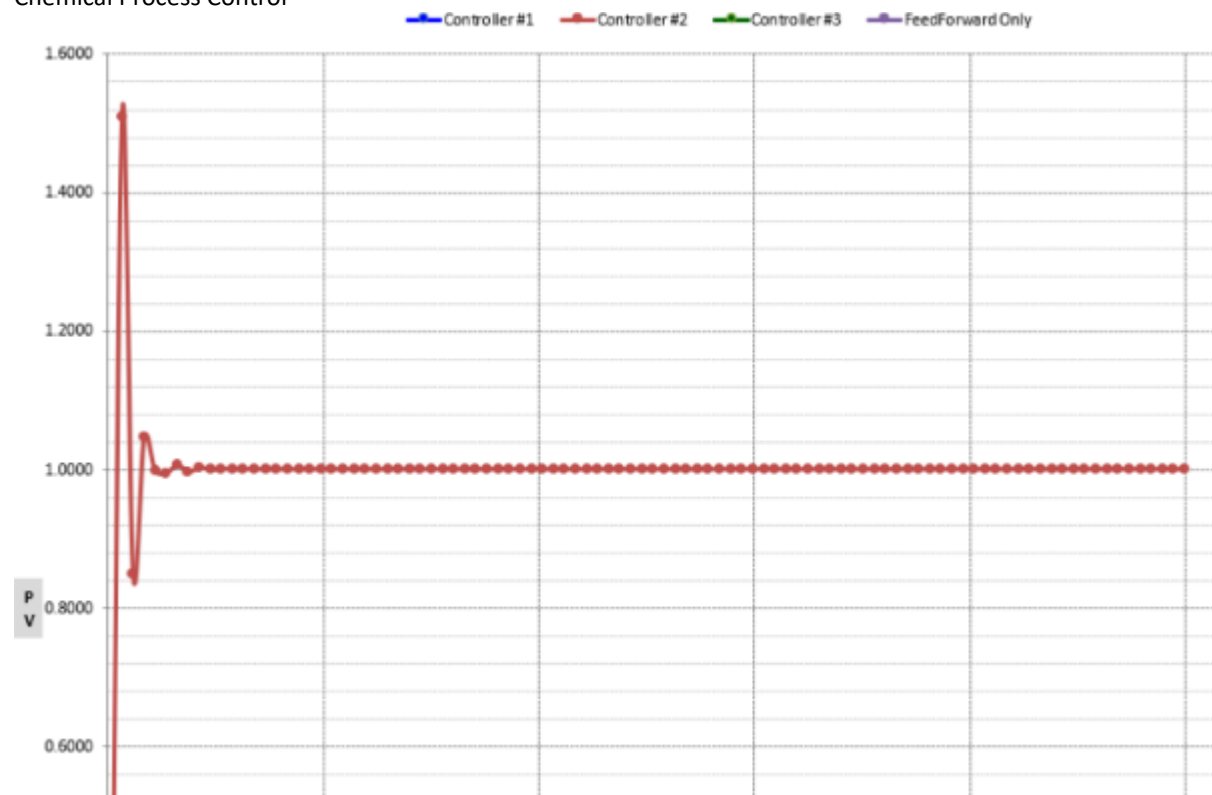
Ziegler-Nichols Open Loop



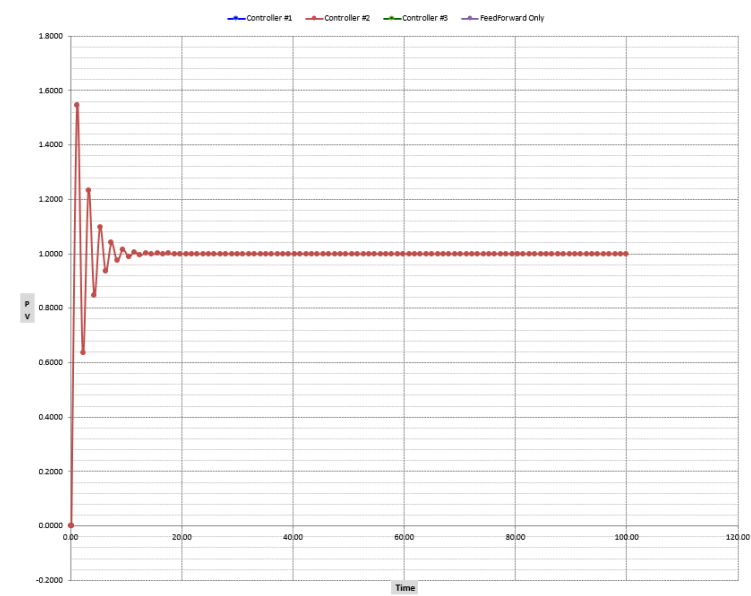
Cohen-Coon



Chemical Process Control

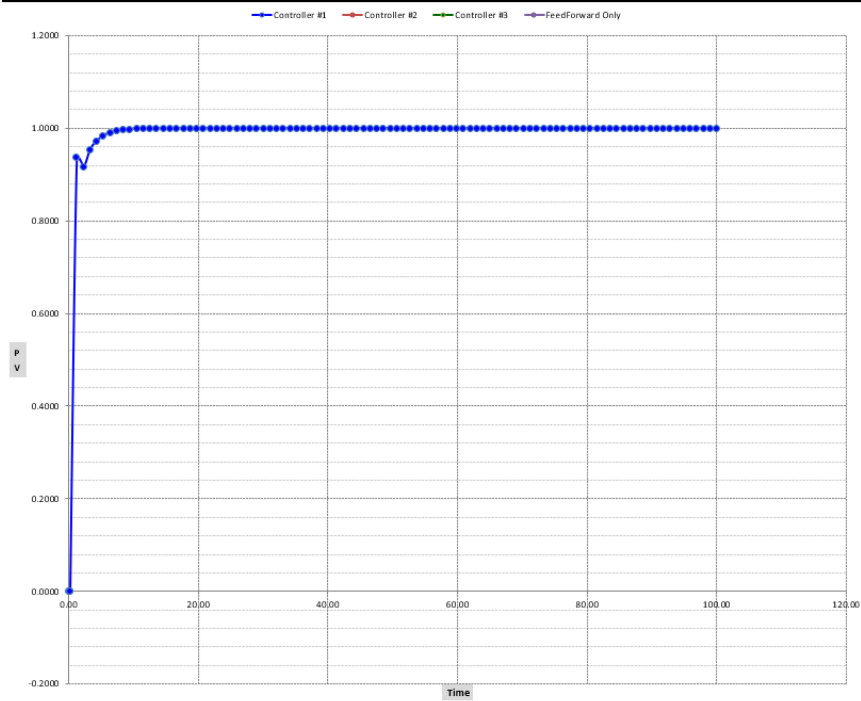


Ziegler-Nichols Closed Loop

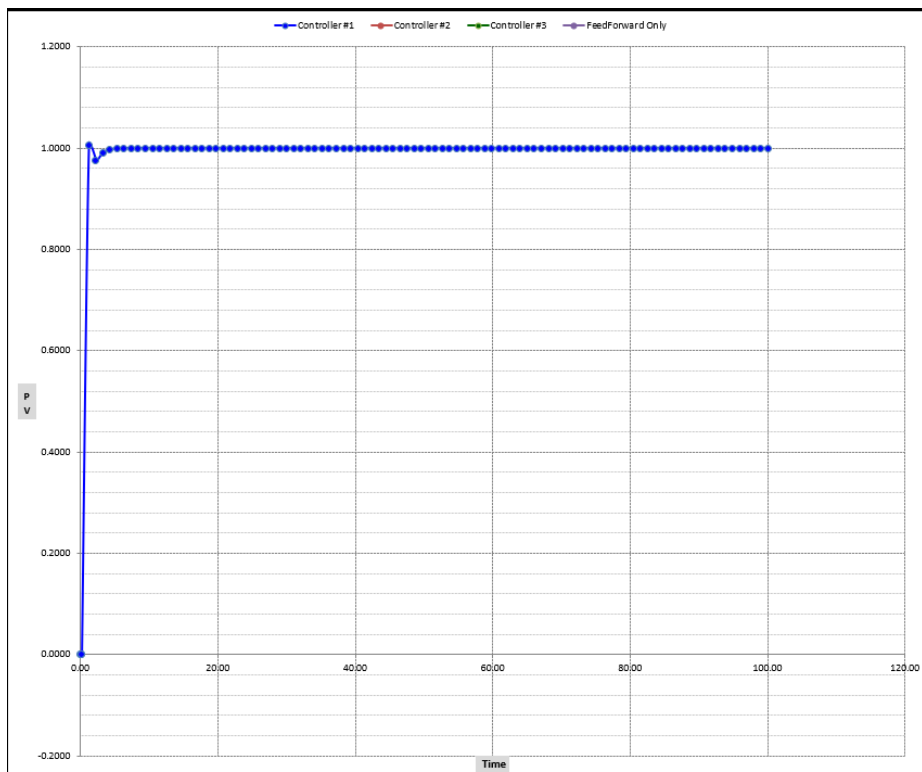


Luyben-Tyreus

Chemical Process Control



IMC



My controller:

Chemical Process Control

